



GAME DEVELOPMENT

Intro to key concepts



WHAT IS GAME DEV?



- **Game development** is the process of creating video games.
- Involves both technical (coding, programming) and creative (design, art, storytelling) elements.
- **Key roles:** Programmers, Designers, Artists, Sound Engineers, Writers.



Game Design Principles



Gameplay Mechanics

Rules and systems (movement, combat, puzzle-solving)



Player Experience

How the game engages players (feedback, difficulty)



Game Objectives

What players aim to achieve
(level completion, high scores)

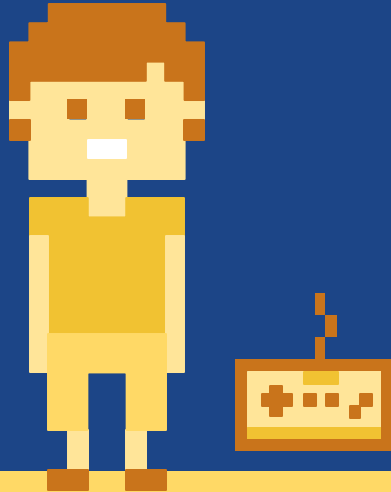


Storytelling

Crafting narratives, characters, and worlds



Programming Concepts



- Game Engines: Unity, Unreal, Godot
- Programming Languages: C# (Unity), C++ (Unreal)
- Essential Coding Skills: Variables, loops, functions (CSCI 1470 C++ / CSCI 3329 Python)
- Event-driven Programming : How games respond to player input (CSCI 3326 Java)
- Object-Oriented Programming: Classes, inheritance, polymorphism (CSCI 3326 Java/3327 OOP)



Graphics and Animation



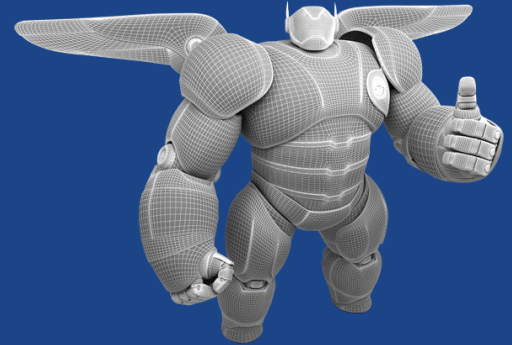
2D Assets

Sprites for characters and environments



3D Models

Textures and meshes for 3D worlds



Animation Basics

Textures and meshes for 3D worlds



Physics and Collision Detection

Basic Physics: Simulating realistic movements (gravity, velocity)

Collision Detection: Detecting when objects interact or collide



✚ User Interface (UI) and User Experience (UX) ✚

UI Design: Menus, heads-up displays (HUD)

UX Design: Feedback, intuitive navigation

Player Interaction: Buttons, sound effects,
visual cues



Sound Design and Music

Sound Effects: Enhancing gameplay with sound

Music: Setting the tone of the game

Voice Acting: Integrating dialogue into the game



Game Testing and Debugging



Playtesting

Testing for bugs and
improving gameplay



Debugging

Finding and fixing issues in
the code



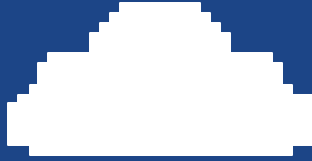
GitHub

Version Control is used to Tracking changes and
managing code in teams



▮ Collaboration and Teamwork ▮





CONCLUSION

Game development is a blend of creativity and technical skills

Continuous learning and collaboration are key

Start small and iterate as you learn more

